

Retrofitting a commercial RF induction generator into a computer-controlled, vacuum-integrated annealing system for reactive-metal grain growth

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Abstract

High-temperature vacuum annealing near a metal's melting point drives controlled grain growth but normally requires expensive turn-key vacuum induction furnaces. We describe an open, generator-agnostic modernization stack that converts a bare commercial RF induction generator into a computer-controlled, vacuum-integrated, pyrometer-feedback annealing system. The contribution is the reproducible control/sensing/vacuum layer — LabVIEW/DAQ analog power control, dual-wavelength optical temperature feedback, a high-vacuum quartz-tube chamber, and an in-house machined graphite crucible/susceptor — which transfers between induction generators provided the generator exposes a monotonic analog power-control input. The reproducible build is anchored on the lab's current, USA-sourced generator (CEIA “Power Cube” 6 kW, procured through East Coast Induction); a ~1970 LEPEL high-frequency furnace served as the originating prototype. The retrofit and consumables add roughly \$1–2k on top of a generator the lab already owns. We report the hardware, design files, bill of materials, build and operating procedures, and a validation grounded in ~100 archived anneal logs: a fixed-geometry power→temperature calibration ($R^2 = 0.991$ over 1200–1400 °C), repeatable 1200 °C / 12 h Ni4N5 processing (1201.2 ± 1.3 °C across 8 runs), stable long-duration control up to 40 h at 1325 °C, and SEM/EBSD/optical microstructural characterization of annealed reactive-metal specimens cross-referenced to their logged thermal histories.

Keywords: induction furnace, grain growth, vacuum annealing, LabVIEW, pyrometer, open-source hardware

Status: working draft generated from the repository's design files, standard operating procedure, parts list, schematics, and run logs. [TODO:]-marked items flag values or figures that still need to be confirmed or added before submission. The reproducible build and bill of materials are anchored on the lab's current, USA-sourced induction generator (CEIA “Power Cube” 6 kW, procured through East Coast Induction); the ~1970 LEPEL furnace is documented only as the originating prototype. An independent review of this draft against the HardwareX section requirements is recorded in `edison-feedback/`.

1. Specifications table

2. Hardware in context

High-temperature annealing near a metal's melting point drives controlled **grain growth**, which is central to studying microstructure–property relationships in metals such as nickel and iron [1, 2, 3]. Doing this without oxidizing the sample requires both high temperature (~1400–1500 °C) and high vacuum (or inert atmosphere) [4, 5]. Commercial vacuum induction annealing furnaces that meet these requirements are expensive turn-key systems, which puts them out of reach for many academic labs.

This work takes the opposite approach: rather than building or buying a complete furnace, we **retrofit and modernize a bare RF induction generator** into a full annealing system by adding (1) computer power control, (2) closed-loop optical temperature control, (3) a high-vacuum quartz-tube chamber,

and (4) a machined graphite crucible/susceptor. The emphasis of the contribution is the **modernization stack itself**, which is deliberately **generator-agnostic**: the same control software, vacuum chamber, pyrometer feedback, crucible, and work-coil geometry transfer from one induction generator to another. The only requirement a reader's generator must satisfy is an **analog power-control input with a monotonic command→power response** over the usable range (e.g. 4–20 mA, 0–5 V, or 0–5 mA); near-linearity is convenient but not required, because the pyrometer feedback and PID calibration absorb any nonlinearity. That portability is the reproducible result and is what makes the design reusable by other labs that already own, or can buy, such an induction generator.

A second feature that broadens the system's applicability is the **machined graphite crucible/susceptor**: because graphite couples strongly to the RF field and heats the charge indirectly by conduction and radiation, the furnace is not limited to electrically conductive samples — it can also process poorly- or non-coupling materials such as YSZ ceramic. This makes the same build useful well beyond reactive-metal annealing.

The modernization was first prototyped on a ~1970 LEPEL high-frequency induction furnace owned by the BYU Mechanical Engineering department — at the time the only departmental machine able to reach the Ni/Fe melting range. A “remote control” capability was added in August 2019 to drive the generator's power from a computer instead of the front-panel knob. The lab subsequently moved the same stack onto a current, USA-sourced solid-state generator (CEIA “Power Cube”

Hardware name	Computer-controlled, vacuum-integrated induction annealing system
Subject area	Engineering and materials science (thermal processing / grain growth)
Hardware type	Heat-treatment / annealing furnace; vacuum system; instrument control
Closest commercial analog	Vacuum induction annealing furnace (turn-key systems cost \$50k–\$200k+)
Open source license	CERN-OHL-S v2 (hardware / design files); MIT (control code)
Cost of hardware	Retrofit + consumables $\approx$ \$1–2k on top of a compatible induction generator; for a from-scratch build the generator + heating head + controller + chiller adds $\approx$ \$16.5k (CEIA quote 210203AP, USD, 2021)
Source file repository	https://doi.org/10.5281/zenodo.20878017 (archived deposit); development repository https://github.com/vertical-cloud-lab/custom-induction-furnace

6 kW via East Coast Induction), whose Power Controller C-V3 Plus accepts a standard analog (4–20 mA) power setpoint. An intermediate, separately-purchased import generator (CYSI GP-15A) was evaluated but did not meet the lab’s control requirements — the required analog remote-control input was lost across vendor model substitutions — and was retired. The reproducible build documented here therefore targets the current USA generator, with the LEPEL retrofit retained as historical context.

Compared with commercial vacuum annealing furnaces, this approach has **substantially lower capital cost** (the retrofit and consumables add roughly \$1–2k on top of a generator a lab already owns, versus \$50k–\$200k+ for turn-key systems), is repairable and modifiable by the user, and is broadly reusable: any lab with an induction generator and an analog power input can reproduce the control, vacuum, and temperature-sensing additions described below.

2.1. Reproducibility boundary

To be explicit about what this paper does and does not reproduce: **this work does not reproduce the RF generator internals**. It reproduces the *modernization layer* built around a compatible generator — the computer power control, optical (pyrometer) feedback, high-vacuum quartz-tube chamber, machined graphite crucible/susceptor, work-coil geometry, and the mechanical support fixturing. A reader supplies (or already owns) a compatible generator and rebuilds the layer described here.

2.2. Prior art and comparison

2.3. Generator compatibility requirements

The control/sensing/vacuum layer transfers to any generator that meets these criteria:

- Exposes an **external analog power-control input** (e.g. 4–20 mA or 0–5 V/0–5 mA).
- The command→delivered-power response is **monotonic over the usable range** (near-linear is convenient but not required, since the pyrometer feedback closes the loop).
- Supports a **water-cooled induction head/coil** matching the documented coil dimensions.
- Delivers **enough RF power** to reach the target temperature for the documented crucible/sample stack.
- Allows **safe electrical isolation** between the DAQ/control path and the generator input.

- Has the required cooling and line power for the selected generator.

When porting to a new generator, the following must be re-established: analog-command scaling, power-to-temperature calibration, PID gains, coil/sample alignment, maximum safe ramp rate, and any generator-specific interlock behavior.

2.4. Transferability: LEPEL prototype → CEIA reproducible build

3. Hardware description

Table 1 lists the technical specifications of the build documented here. These are *build-dependent* values (a reader supplies their own generator, pump, and pyrometer) and so are kept out of the mandatory HardwareX specifications table.

The system comprises five subsystems (see the schematics in `extracted-context/schematic_induction_furnace.md` and `extracted-context/schematic_support_stand.md`):

1. **RF generator + work coil.** A solid-state induction generator (current build: CEIA Power Cube PW3-90/50) drives a water-cooled copper work coil. The coil carries both the oscillating current and cooling water. Coil geometry ($\sim$ 3 in tall, 2.5 in ID, $\sim$ 6.5 turns) was tuned to the sample/crucible stack [6] (see `extracted-context/order_corrections_coils.md`).
2. **Vacuum + gas.** A quartz tube forms the chamber and is connected through a KF40 flange and flexible bellows to an Edwards T-Station 85 turbo-molecular pump ($\sim$ 1 $\times$ 10 $^{-6}$ –10 $^{-8}$ Torr). A manual vent valve and an inert-gas regulator allow controlled venting/backfilling to suppress oxidation of reactive metals. The vacuum connection stack, in order, is: **quartz tube → KF40 flange + optical window → flexible bellows → vacuum gauge → Edwards T-Station 85 → vent / inert-gas backfill branch**.
3. **Optical temperature sensing.** A dual-wavelength (ratio) pyrometer (the canonical unit is a LumaSense IMPAC ISR 6, 800–2500 °C; the IMPAC brand is LumaSense’s) views the sample through a KF40 optical window, providing a non-contact temperature signal that is robust to emissivity changes and to the RF field that would corrupt thermocouples [7, 8, 9].
4. **Control / DAQ + LabVIEW.** A LabVIEW DAQ analog output sets generator power; the pyrometer signal is

Class	Examples	Limitation this design addresses
Commercial vacuum induction annealing furnaces	turn-key systems (\$50k–\$200k+)	Cost; closed/non-modifiable; vendor lock-in
Open non-induction annealing systems	tube/box furnaces with resistive heating	Lower max temperature and slower ramp than induction; no RF-susceptor heating of non-coupling charges
Induction generators sold bare	CEIA, Ambrell, etc.	Ship without integrated vacuum, optical feedback, or computer setpoint control — exactly the layer added here

Aspect	LEPEL prototype (~1970)	CEIA reproducible build (current)	Transferred?
Analog control interface	0–5 mA	4–20 mA (Power Controller C-V3 Plus)	Concept yes; scaling re-set
Frequency	~200–450 kHz tube	solid-state (generator-set)	n/a (generator internal)
Max power	up to ~35 kW	6 kW	Re-tuned for charge
Pyrometer / optical feedback	dual-wavelength	same dual-wavelength stack	Yes
Vacuum chamber / quartz tube	quartz + KF40	same	Yes
Graphite crucible/susceptor	machined graphite	same	Yes
Support stand / fixturing	bolted stand	same	Yes
LabVIEW/DAQ control concept	manual + ramp/PID VIs	same VIs	Yes

read back for closed-loop ramp/soak control. The VIs implement manual control, automated ramping, PID tuning, and email alerts (see `./code/`). In the CEIA build the DAQ outputs **0–5 V** to a **voltage-to-current loop conditioner** that converts the command to the controller's **4–20 mA** power-setpoint input (the LEPEL prototype instead took a 0–5 mA command directly). The closed-loop **signal flow** is therefore: **LabVIEW setpoint** → **DAQ analog output (0–5 V)** → **V→I conditioner (4–20 mA)** → **generator power** → **sample/crucible heating** → **pyrometer optical reading** → **DAQ analog input (scaled to °C)** → **PID controller** → **updated analog command**. The same path runs **open-loop** (direct command ramp) while the temperature is below the pyrometer's ~700 °C detection floor; once the pyrometer reports a **sustained valid reading above that floor**, control hands over to **closed-loop PID** on the optical temperature [10].

- Mechanical support stand + crucible.** A bolted support stand (8 short + 4 long sections, 12 corner braces, 4 floor mounts) positions the chamber, coil, pyrometer holder, and quartz disc. An **in-house machined graphite crucible** holds the sample and acts as the RF susceptor.

The LabVIEW control front panel (Fig. 1) exposes the ramp-up, PID-soak, and cool-down setpoint tables, the PID gains, the inert-gas flow control, and live power-command and pyrometer-temperature traces used to run and monitor an anneal.

3.1. Custom-machined graphite crucible / susceptor

Because some charges of interest couple poorly to the RF field (e.g. YSZ ceramic) or must be physically contained, the sample is held in a **graphite crucible machined in-house from 3000 °C-grade graphite stock** (Cotronics 56L-3). The graphite couples strongly to the induction field and acts as a

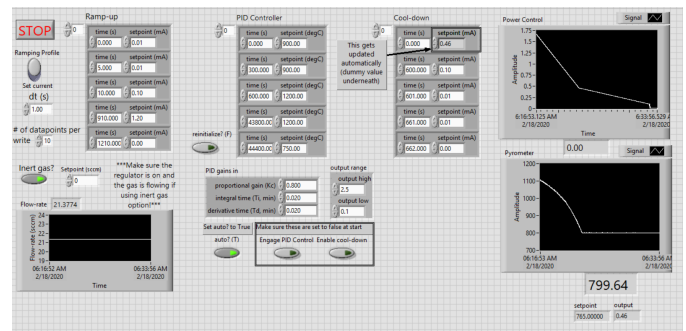


Figure 1: LabVIEW control front panel (docs/equipment-reference/frontPanel1.200219.PNG): editable ramp-up / PID-soak / cool-down setpoint tables (time vs. analog command in mA below the pyrometer floor and vs. °C once closed-loop), PID gains and output range, inert-gas flow setpoint/readback, and live power-command and pyrometer-temperature strip charts. Here the pyrometer reads ~800 °C during ramp.

susceptor, transferring heat to the contained charge by conduction/radiation; this makes the same furnace usable for conductive metals (direct coupling) and for non/poorly-coupling materials (indirect, susceptor-driven heating). Crucibles are machined to fit inside the quartz tube and seat on a ceramic-cylinder / Teflon spacer stack. Cracked crucibles are repaired with a 3000 °C graphite cement (Resbond 931) rather than discarded. **[TODO: add the crucible machining drawing and photos to paper/figures/.]**

3.2. Who can use this system

- Materials labs needing **oxidation-free grain-growth anneals** of reactive metals (Ni, Fe) near their melting points.
- Labs that own a **bare induction generator** and want to add computer setpoint control, optical feedback, and vac-

Specification	Value (this build)
Induction generator (reproducible build)	CEIA “Power Cube” PW3-90/50 solid-state RF generator + Power Controller C-V3 Plus (East Coast Induction, USA)
Induction generator (prototype)	LEPEL high-frequency tube furnace (~1970), up to ~35 kW
Power-control input	4–20 mA analog setpoint via LabVIEW/DAQ → V→I conditioner (CEIA build; the LEPEL prototype used 0–5 mA — see migration table)
Generator-input requirement	External analog power-control input with monotonic command→power response over the usable range
Control signal chain	DAQ analog output (0–5 V) → voltage-to-current (4–20 mA) loop conditioner → generator power-setpoint input; pyrometer analog output → DAQ analog input (scaled to °C)
Max sample temperature	~1400–1500 °C+ (Ni/Fe melting range)
Pyrometer detection range	~800–2500 °C (dual-wavelength ratio); does not report below ~700 °C
Demonstrated soak duration	minutes up to ~40 h (per docs/data_log/ run logs)
Sample / crucible envelope	charge fits inside a 35 mm-ID quartz tube and a machined graphite crucible
Operating vacuum	~1×10 ⁻⁶ to 1×10 ⁻⁸ Torr (turbo-molecular pump)
Temperature measurement	Dual-wavelength ratio pyrometer (LumaSense IMPAC ISR 6, ~800–2500 °C)
Chamber	Quartz tube, KF40 optical-window flange, inert-gas vent
Crucible / susceptor	In-house machined graphite (3000 °C grade) crucible-susceptor
Work coil	Water-cooled copper, ~3 in tall, 2.5 in ID, ~6.5 turns

Table 1: Technical specifications of the documented build. Build-dependent values that a reader replaces with their own generator/pump/pyrometer; not part of the mandatory HardwareX specifications table.

uum integration without buying a turn-key furnace.

- Groups processing **poorly- or non-coupling charges** (e.g. YSZ ceramic) that can be heated indirectly through the graphite susceptor.
- Teaching/maker contexts wanting a **repairable, modifiable** high-temperature vacuum furnace built from documented, open design files.

4. Design files summary

[TODO: export the LabVIEW .vi block diagrams to PDF/PNG so non-LabVIEW readers can inspect the control logic.]

5. Bill of materials summary

HardwareX requires a single 7-column BOM table. Groupings are denoted by the designator prefix: **GEN** = induction generator + controller (the current USA reproducible build; the legacy LEPEL is the prototype alternative, not a line item), **RETRO** = the reproducible retrofit (control / vacuum / sensing) parts, and **CONS** = consumables. Full vendor/price detail for the RETRO and CONS parts is in extracted-context/parts_list.md (extracted from docs/induction_parts_list.xlsx).

Costing conventions: all costs are in USD; prices are the lab’s purchase records (quote year ~2019–2021), several from used/surplus/eBay sources as noted; shipping and tax are excluded unless stated. **The build instructions and the BOM below are canonical for the CEIA system only;** the legacy LEPEL prototype and the rejected CYSI import path are documented in the text and the transferability table, not in the BOM.

[TODO: confirm RETRO-1/4/5 pricing, the exact NI DAQ model, and the V→I loop-conditioner model. GEN

line items are filled from East Coast Induction quote 210203AP (3 Feb 2021; generator + heating head + controller + chiller + line transformer total \$16,487).]

6. Build instructions

1. **Support stand.** Assemble the bolted stand (8 short + 4 long sections, 12 corner braces, 4 floor mounts) and mount the generator/heating head and pyrometer holder.
2. **Work coil.** Form/water-cool the copper coil to the corrected geometry (~3 in tall, 2.5 in ID, ~6.5 turns); connect to the generator head and the cooling-water circuit. Use the before/after figures from induction_order_corrections.pptx as the reference. **[TODO: state the exact copper-tube OD/wall, bend radius/spacing method, and cooling-connection fittings from the coil drawing (docs/coils-drawing.pdf).]**
3. **Graphite crucible/susceptor.** Machine the crucible from 3000 °C graphite stock to fit inside the quartz tube and accept the sample. Seat it on the ceramic-cylinder / Teflon spacer stack. Repair any cracks with graphite cement.
4. **Vacuum chamber.** Mount the quartz tube with the KF40 flange and optical window. The **reference build** seats a 55 mm quartz window on the overpressure KF40 centering ring and secures it with the plastic quick-clamp (reusable, no adhesive); a Torr-Seal’d blank flange is documented as an alternative for a permanent seal. Connect the flexible bellows to the Edwards T-Station 85 turbo pump; install the manual vent valve and inert-gas line.
5. **Control wiring.** Wire the DAQ analog output (0–5 V) through the voltage-to-current loop conditioner to the generator’s analog power-control input (4–20 mA on the current CEIA controller; the LEPEL prototype took 0–

Design file	File type	License	Location
LabVIEW control VIs (furnace control, manual control, PID tuning v1/v2, email alert)	LabVIEW .vi	MIT (proposed)	../code/induction-furnace-control-code/
Manual ramping v5 VIs	LabVIEW .vi	MIT (proposed)	../code/manual-ramping-v5/
plotheatcurve.m (heat-curve plotter)	MATLAB	MIT (proposed)	../code/plotheatcurve.m
Work-coil drawing	PDF / OXPS	CERN-OHL-S (proposed)	docs/coils-drawing.pdf, docs/coils.oxps
System schematics	PPTX (+ extracted text/figures)	CERN-OHL-S (proposed)	docs/*.pptx, extracted-context/
Temperature-control wiring	PNG	CERN-OHL-S (proposed)	docs/temp-control-modification/0-5V PLC wire design.png
KF40 overpressure centering ring	STEP	CERN-OHL-S (proposed)	docs/KF Supplies/KF40-overpressureCenteringRing.step
Graphite crucible/susceptor machining drawing	[TODO: add]	CERN-OHL-S (proposed)	[TODO:]

5 mA directly). Wire the pyrometer output into the DAQ analog input and, for closed-loop ramping, the pyrometer/PLC temperature-control path (docs/temp-control-modification/). Maintain electrical isolation between the DAQ/control path and the generator interface.

6. **Software.** Deploy the LabVIEW VIs from ../code/ and configure the DAQ channels; verify manual control before enabling automated ramp/soak.

[TODO: curate a clean, photographed build sequence — candidate photos exist in docs/data_log/.../IFrun036_* and the PPTX media.]

7. Operation instructions

Condensed from the lab SOP (extracted-context/SOP_induction_200901.md).

7.1. Preflight checklist (before energizing RF)

- ☐ Chamber at high vacuum ($\sim 1 \times 10^{-6}$ Torr or better) or backfilled to the intended inert atmosphere.
- ☐ Cooling-water flow confirmed at the coil/generator (no leaks).
- ☐ Coil/sample alignment and chamber clearance verified.
- ☐ Pyrometer aligned and sighted on the sample/crucible.
- ☐ No conductive objects (jewelry, watches, phones, tools) within ≥ 6 –12 in of the coil.

7.2. Startup

1. **Load the sample.** With the chamber vented to atmosphere (never open the vent valve while the turbo pump is spinning), stack sample → optional alumina plate → ceramic cylinder → Teflon spacer inside the graphite crucible, insert into the quartz tube, and clamp the KF40 flange finger-tight with the O-ring.
2. **Pump down.** Close the vent valve, start the roughing/turbo pumps, and wait for high vacuum ($\sim 1 \times 10^{-6}$ Torr or better).

3. **Cooling water.** Check for leaks and confirm flow before applying RF power.

7.3. Normal run

4. **Power on.** Bring up the generator per its sequence, then run the LabVIEW ramp/soak profile. Below the pyrometer's $\sim 700^\circ\text{C}$ detection floor the VI ramps **open-loop** on the analog command; once the pyrometer returns a **sustained valid reading above the $\sim 700^\circ\text{C}$ floor** (the handoff criterion), control switches to **closed-loop** PID on the optical temperature.
5. **Anneal.** Hold the temperature/time profile (e.g. $\sim 1200^\circ\text{C}$ for several hours for Ni grain growth); the VI can email status alerts.

7.4. Shutdown

6. **Ramp down.** Ramp power down and stop RF, then turn off cooling water, allow the sample to cool under vacuum, and vent only after the turbo pump has fully stopped.

7.5. Abort / emergency stop

- **Loss of cooling water or chiller alarm:** immediately command power to zero / stop RF, then investigate; do not re-energize until flow is restored.
- **Vacuum loss or quartz-tube fracture:** stop RF immediately; do not vent against a spinning turbo pump.
- **Arcing, smoke, or runaway temperature:** zero the analog command, kill RF at the generator, and lock out before inspecting.

8. Validation and characterization

The repository contains 100+ logged runs (docs/data_log/, IFrun001–100) across Ni200, Ni4N5, YSZ, and Pd thermal-evaporation experiments, with temperatures of ~ 900 – 1400°C and soak durations from minutes to ~ 40 h, each with .xlsx/.txt/.lvm traces and some photos. Per-run

Desig.	Component	No.	Unit \$	Total	Source	Material
GEN-1	CEIA "Power Cube" PW3-90/50 solid-state RF generator (V3000-0070)	1	\$8,240	\$8,240	East Coast Induction (USA)	—
GEN-2	Power Controller C-V3 Plus (V3000-0414)	1	\$1,670	\$1,670	East Coast Induction (USA)	—
GEN-3	Recirculating water chiller (MIL043008 air-cooled closed-loop, V1650-0060)	1	\$1,575	\$1,575	East Coast Induction (USA)	—
GEN-4	Heating head + 3 m cable and water lines (PWH-13-12-30/50, V3000-0300); 1 coil made to spec at no charge with a complete unit	1	\$3,337	\$3,337	East Coast Induction (USA)	copper
GEN-5	Line auto-transformer, 12 kVA / 3PH / 480×380 V with taps (V049-0508; facility-dependent)	1	\$1,665	\$1,665	East Coast Induction (USA)	—
RETRO-1	LabVIEW-compatible DAQ with analog out + in (e.g. NI USB-6001/6009-class, 0–5 V AO) [TODO: confirm exact model]	1	[TODO:]	[TODO:]	National Instruments	—
RETRO-2	Dual-wavelength ratio pyrometer (LumaSense IMPAC ISR 6, 800–2500 °C)	1	\$241 (used)	\$241	eBay — used (new list ~\$5,500, LumaSense quote 00161403)	—
RETRO-3	24 V linear power supply for pyrometer (International Power)	1	\$51	\$51	Mouser	—
RETRO-4	Voltage-to-current (0–5 V → 4–20 mA) loop conditioner for the generator power-setpoint input	1	[TODO:]	[TODO:]	[TODO: signal-conditioning vendor]	—
RETRO-5	Edwards T-Station 85 turbo-molecular pump + gauge	1	[TODO:]	[TODO:]	Edwards	—
RETRO-6	Edwards TAV5 vent valve	1	\$59 (used)	\$59	eBay — used (price from purchase record; verify current listing or use a vacuum-supply quote)	—
RETRO-7	Inert-gas regulator (Fisher FS-50)	1	\$38	\$38	Fisher Scientific	—
RETRO-8	KF40 overpressure centering ring	1	\$19	\$19	IdealVac	aluminum
RETRO-9	KF40 plastic quick vacuum clamp	1	\$23	\$23	IdealVac	polymer
RETRO-10	Optical window, custom quartz disc 55 mm × 1.5 mm (1 used; buy 3 for spares)	3	\$18 (est.)	\$54	Custom-cut metric disc — vendor not recorded in parts list; price approximate (McMaster stocks only imperial sizes)	fused quartz
RETRO-11	Ultra-high-temperature quartz disc 2 in × 1/16 in (McMaster 1357T12)	1	\$19	\$19	McMaster-Carr	fused quartz
RETRO-12	Inert-gas plumbing (BSPT/KF adapters, barbs, tee, 10 psi relief valve McMaster 4772K4)	1 set	see parts list	~\$66	McMaster-Carr / BMotionTech	brass / steel
CONS-1	Quartz tube, 4 ft × 35 mm ID, cut to length	2	\$67	\$134	QSI Quartz	fused quartz
CONS-2	Graphite stock for machined crucible/susceptor (56L-3, 3000 °C, 1×6×6 in)	1	\$99	\$99	Cotronics	graphite
CONS-3	Graphite repair cement (Resbond 931-1, 3000 °C)	1	\$108	\$108	Cotronics	graphite
CONS-4	Alumina crucible stock (RTC-60-2, 1787 °C, 10 lb)	1	\$91	\$91	Cotronics	alumina
CONS-5	Zirconia crucible stock (760-1, 2204 °C, 10 lb)	1	\$124	\$124	Cotronics	zirconia
CONS-6	Torr-Seal high-vacuum epoxy / O-rings	1 set	\$57	\$57	Varian / McMaster	epoxy / elastomer

traces were parsed into uniform CSV/PNG files (docs/data_log/processed/; build_run_traces.py), and the quantitative metrics below were computed reproducibly from those traces by paper/build_validation_figures.py (values mirrored in paper/validation-metrics.json). Because the archive is a retrospective operating record rather than a prospectively balanced design of experiments, every comparison is restricted to an internally consistent subset (matching material, nominal profile, and operating configuration) and the power→temperature relation is reported as an *operational* calibration for one fixed geometry, not as a universal transfer function.

System overview (Fig. 2). Figure 2 is a labeled schematic of the full system — recirculating water chiller, induction generator, heating head, water-cooled work coil, quartz-tube vacuum chamber with optical quartz disc, pyrometer holder/pyrometer line-of-sight, pressure sensor, bellows, and the Edwards T-Station 85 turbo-molecular pump on the bolted support stand, with the DAQ/control laptop closing the loop. (Hardware photographs are collected in paper/extracted-context/figures/; a labeled composite photograph will replace this schematic on final submission.)

Power-command → temperature calibration (Fig. 3). For one fixed Ni4N5/quartz/graphite/optical configuration, four archived steady-state anneals spanning 1200–1400 °C (IFrun039, IFrun040, IFrun038, IFrun032) give a monotonic relation between the soak-mean analog power command and the soak-mean pyrometer temperature. A linear guide-to-the-eye fit over these points is $T \approx 931 + 632 I$ (°C, I in mA), $R^2 = 0.991$, $n = 4$. This is presented as an operational calibration only: the broader archive shows configuration-dependent offsets (e.g. runs near 1200 °C required soak-mean commands of ~0.44 mA vs. ~0.66 mA in a different era/geometry), so the fit must not be read as a universal command–temperature law.

Representative ramp/soak with control metrics (Fig. 4). Run IFrun079_Ni4N5_084_1300C_12h is a representative closed-loop anneal. Measured against the 1300 °C setpoint, the soak held 1302.1 ± 3.0 °C (mean ± SD) over the 12 h plateau, with a 10–90 % rise time of ~1135 s, an overshoot of ~12 °C, a maximum in-soak deviation of ~10 °C, and a near-zero linear drift of ~0.02 °C/h, demonstrating that the pyrometer-feedback loop holds a programmed soak over many hours.

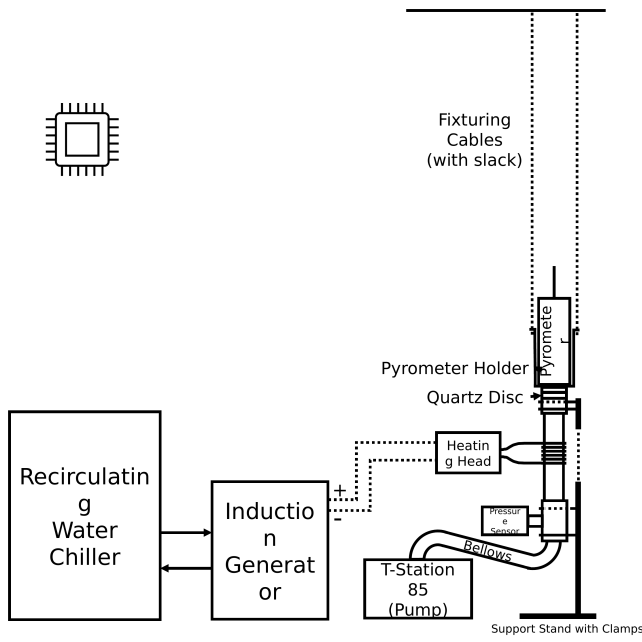


Figure 2: System overview of the computer-controlled, vacuum-integrated induction annealing system: recirculating water chiller and induction generator drive a water-cooled heating head/work coil around a quartz-tube vacuum chamber; a dual-wavelength pyrometer sights the charge through a quartz optical disc; an Edwards T-Station 85 turbo-molecular pump (with pressure gauge and bellows) evacuates the chamber; and a DAQ/laptop sets generator power and reads the pyrometer for closed-loop control. Rendered from the editable source schematic docs/induction-furnace-schematic.pptx.

Repeatability and long-soak stability (Figs. 5–6). Run-to-run repeatability was assessed on the most internally consistent cohort available: eight Ni4N5 anneals at a nominal 1200 °C / 12 h condition (IFrun052–IFrun060, excluding missing run numbers), all logged under the same gas-flow configuration. Across these runs the soak temperature was 1201.2 ± 1.3 °C (range 1199.8–1204.2 °C; coefficient of variation 0.11 %; $n = 8$), peak temperature 1212.6 ± 6.7 °C, and soak duration 766 ± 15 min. Runs from other periods or geometries (e.g. IFrun016, IFrun019, IFrun049) are deliberately *not* pooled into this estimate. Long-duration stability was demonstrated with two Ni200 anneals at 1325 °C: a 20 h run (IFrun081) held 1326.5 ± 1.5 °C with drift -0.004 °C/h, and a 40 h run (IFrun080) maintained ~ 1326 – 1329 °C over 2400 min in-band.

Microstructural validation (SEM + EBSD + optical). The furnace exists to change microstructure, so the key validation is **real microstructural data** tied to specific thermal histories. The committed SEM/EBSD and optical-microscopy archives (docs/SEM/, docs/optical/) are keyed by Ni4N5-### / Ni200-### specimen IDs that map directly to the run IDs in docs/data_log/. This mapping is built reproducibly by paper/build_characterization_crossref.py, which parses the specimen ID out of every characterization folder, joins it to the machine-parsed run summary, and writes the per-specimen cross-reference in paper/characterization_crossref.csv/.md: **35 characterized specimens**, of which

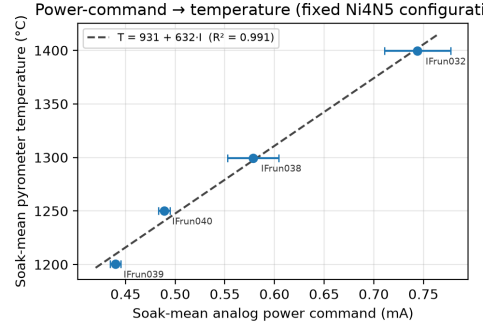


Figure 3: Power-command → temperature calibration for one fixed archived Ni4N5 configuration. Points are soak-mean values; horizontal bars are the in-soak SD of the analog command. The dashed line is a guide-to-the-eye linear fit ($R^2 = 0.991$, $n = 4$).

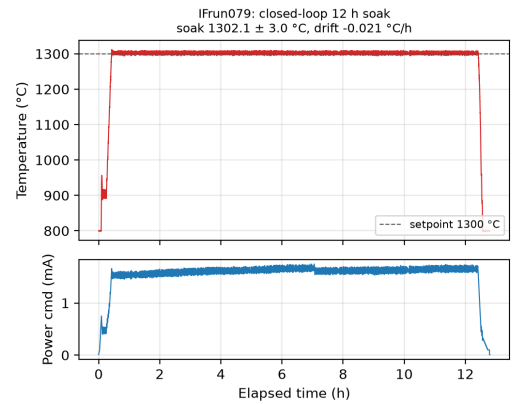


Figure 4: Representative closed-loop anneal (IFrun079, Ni4N5, nominal 1300 °C / 12 h): pyrometer temperature with setpoint overlay (top) and analog power command (bottom). Soak mean 1302.1 ± 3.0 °C, drift -0.021 °C/h.

9 link to a parsed run log, 8 carry committed EBSD/OIM data, and one carries EDS. Inverse-pole-figure (IPF) EBSD orientation maps of two annealed specimens (Fig. 7) resolve hundreds of indexed grains, and the best-correlated specimen, Ni4N5_081 (annealed in IFrun082 at a nominal 1300 °C for 20 h; logged soak ~ 1301 °C), is documented across length scales — optical grain map with in-image grain-size measurements, SEM survey, and high-magnification grain-boundary detail (Fig. 8). The full multi-gigabyte raster/EBSD archive (including .oim, .osc, .ang, and DREAM.3D volumes) remains on the lab’s Box share and is inventoried per file in docs/SEM/CATALOG.csv and docs/optical/CATALOG.csv. Grain size is quantified with: number of grains analyzed, equivalent-circle-diameter or intercept metric, median + IQR (or mean $\pm$ SD), segmentation/cleanup criteria, minimum-grain-area threshold, and whether twins are merged or separated, with the specimen/run (not the individual grain) treated as the experimental unit [1, 3]. Where as-received baselines are unavailable for a batch, results are framed as final-state comparisons across archived anneals rather than as fully parameterized grain-growth kinetics.

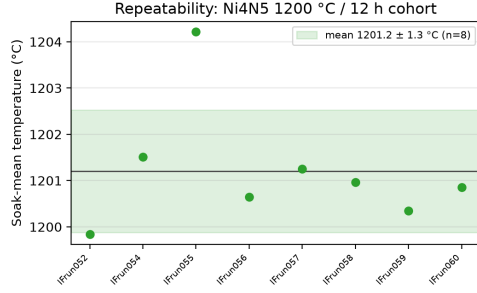


Figure 5: Run-to-run repeatability of the Ni4N5 1200 °C / 12h cohort (IFrun052–IFrun060, $n = 8$). Points are per-run soak-mean temperatures; the band is mean $\pm$ SD (1201.2 ± 1.3 °C).

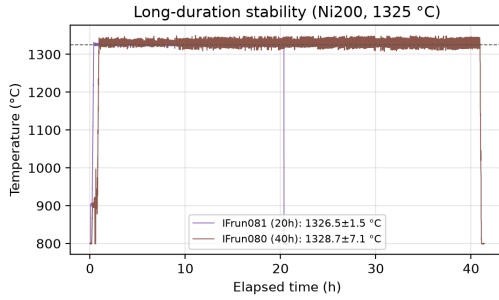


Figure 6: Long-duration thermal stability: Ni200 anneals at 1325 °C held for 20 h (IFrun081) and 40 h (IFrun080).

Suggested controls. that build reviewer trust: open-loop vs. closed-loop on the same target cycle; vacuum vs. inert-gas backfill mode; directly-coupled metal vs. susceptor-mediated ceramic heating; and long-soak (multi-hour) stability.

Specimen $\leftrightarrow$ thermal-history linkage. Each microstructure specimen is tied to its exact run via the table below. “Nominal soak” is the target parsed from the run filename; “logged soak T” is the machine-parsed soak-mean from the trace (run_summary.csv); “atmosphere” indicates whether active gas flow was present in the parsed log summary (has_flow), not a fully reconstructed gas chemistry.

Characterization $\leftrightarrow$ run inventory. Table 2 lists the specimens that carry committed SEM/EBSD/optical data and a matched parsed run log (generated by paper/build_characterization_crossref.py). Two specimens (Ni4N5_034, Ni4N5_081) carry committed EBSD orientation maps on top of SEM/optical imaging, and Ni4N5_081 is the most completely characterized run (Fig. 8). The complete 35-specimen cross-reference, including characterized specimens whose run logs are not in the parsed set, is in paper/characterization_crossref.csv.

8.1. Limitations of the retrospective validation

The validation dataset is retrospective rather than prospectively designed, so the archived runs are not treated as a balanced factorial experiment and comparisons are restricted to

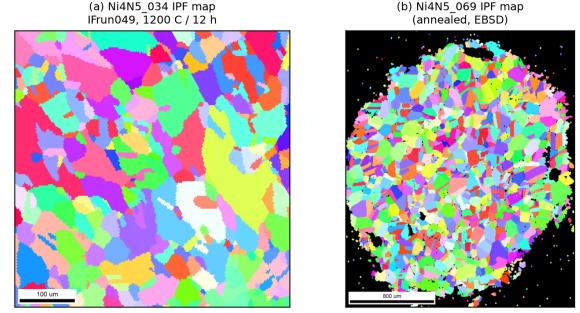


Figure 7: EBSD inverse-pole-figure (IPF) orientation maps of two annealed Ni4N5 specimens, rendered from the committed docs/SEM/ archive. (a) Ni4N5_034 (IFrun049, 1200 °C / 12 h); (b) Ni4N5_069. Each indexed color is a crystallographic orientation; the maps resolve hundreds of grains for grain-size and texture analysis.

internally consistent subsets. Because induction coupling, radiative view factors, and pyrometer line-of-sight depend on specimen/crucible geometry and optical alignment, the power-command/temperature calibration is reported only for a fixed archived configuration and is not a universal transfer function. Reported temperatures are closed-loop ratio-pyrometer readings in a fixed optical geometry; absolute specimen temperature may still depend on emissivity and surface condition not fully reconstructable for every archived run [7, 8]. Run-to-run repeatability is evaluated only within the most consistent cohort (IFrun052–IFrun060); most other temperature/time conditions are singletons, so inference is limited to descriptive summaries. Microstructural comparisons are reported only for specimens whose image files link unambiguously to a logged thermal history; ambiguously labeled images are excluded rather than inferred.

9. Safety

The system combines high voltage/RF, high vacuum, high temperature, and water cooling; the principal hazards and mitigations are summarized below.

PPE / shielding. Operators should wear safety glasses (quartz-fracture/implosion risk), clean nitrile gloves when handling vacuum components, and remove all conductive personal items before approaching the energized coil; keep sensitive instrumentation outside the RF near-field.

10. Ethics statements

This work did not involve human participants, human data, or animal subjects. There are no associated ethical considerations.

Declarations

CRedit author roles. [TODO: confirm] — provisional: Sterling G. Baird (conceptualization, software, investigation, writing); Kevin Cole (methodology, resources, supervision); additional authors to be confirmed.

Ni4N5_081 microstructure after IFrun082 (1300 °C / 20 h, flow)

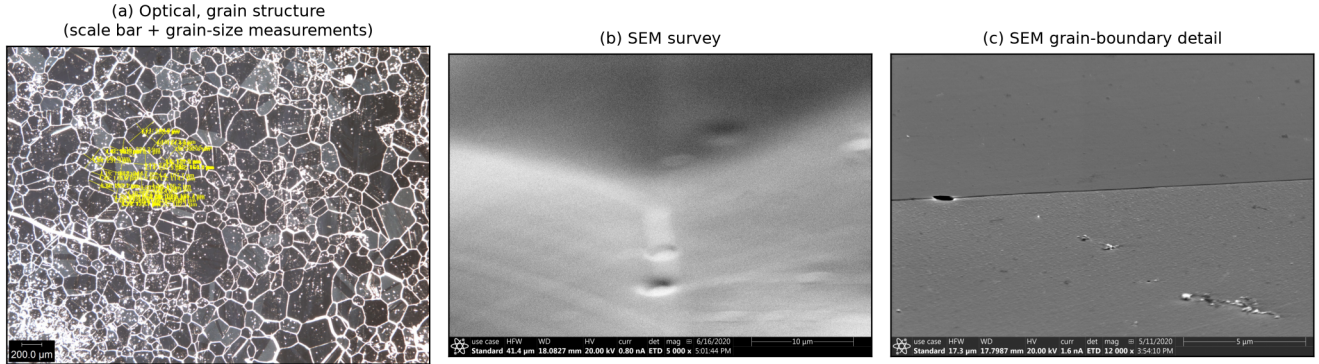


Figure 8: Multi-scale microstructure of specimen Ni4N5_081 after IFrun082 (nominal 1300 °C / 20 h, gas flow), correlating the characterization archive with a single logged thermal history. (a) Optical micrograph (grain structure, 200 µm scale bar, with in-image grain-size measurements); (b) SEM survey; (c) high-magnification SEM grain-boundary detail. All panels are committed files under docs/optical/ and docs/SEM/.

Specimen	Material	Run ID	Nominal soak	Logged soak T	Soak time	Atmosphere	Role / figure
Ni4N5_026	Ni4N5	IFrun039_Ni4N5_026_1200C_6h	1200 °C	1200.4 °C	6 h	no-flow	Calibration (Fig. 3)
Ni4N5_027	Ni4N5	IFrun040_Ni4N5_027_1250C_6h	1250 °C	1250.0 °C	6 h	no-flow	Calibration
Ni4N5_025	Ni4N5	IFrun038_Ni4N5_025_1300C_1h	1300 °C	1299.9 °C	1 h	no-flow	Calibration
Ni4N5_022	Ni4N5	IFrun032_Ni4N5_022_1400C_10min	1400 °C	1399.6 °C	10 min	no-flow	Calibration
Ni4N5_084	Ni4N5	IFrun079_Ni4N5_084_1300C_12h	1300 °C	1302.1 °C	12 h	flow	Representative (Fig. 4)
Ni4N5_040	Ni4N5	IFrun052_Ni4N5_040_1200C_12h	1200 °C	1199.8 °C	12 h	flow	Repeatability (Fig. 5)
Ni4N5_042,044	Ni4N5	IFrun054_Ni4N5_042,044_1200C_12h	1200 °C	1201.5 °C	12 h	flow	Repeatability
Ni4N5_045,046	Ni4N5	IFrun055_Ni4N5_045,046_1200C_12h	1200 °C	1204.2 °C	12 h	flow	Repeatability
Ni4N5_047,048	Ni4N5	IFrun056_Ni4N5_047,048_1200C_12h	1200 °C	1200.6 °C	12 h	flow	Repeatability
Ni4N5_049,050	Ni4N5	IFrun057_Ni4N5_049,050_1200C_12h	1200 °C	1201.2 °C	12 h	flow	Repeatability
Ni4N5_051,052	Ni4N5	IFrun058_Ni4N5_051,052_1200C_12h	1200 °C	1201.0 °C	12 h	flow	Repeatability
Ni4N5_053,054	Ni4N5	IFrun059_Ni4N5_053,054_1200C_12h	1200 °C	1200.3 °C	12 h	flow	Repeatability
Ni4N5_056,057	Ni4N5	IFrun060_Ni4N5_056,057_1200C_12h	1200 °C	1200.9 °C	12 h	flow	Repeatability
Ni200_015	Ni200	IFrun081_Ni200_015_1325C_20h	1325 °C	1326.5 °C	20 h	flow	Long soak (Fig. 6)
Ni200_017	Ni200	IFrun080_Ni200_017_1325C_40h	1325 °C	1326.3 °C	40 h	flow	Long soak
Ni4N5_079,080	Ni4N5	IFrun072_Ni4N5_079,080_1400C_12h	1400 °C	—	12 h	flow	Process-window specimen

Specimen	Material	Matched run	SEM	Opt.	EBSD
Ni200_001	Ni200	IFrun004/005	0	3	no
Ni200_015	Ni200	IFrun081 (1325 °C/20 h)	0	3	no
Ni200_017	Ni200	IFrun080 (1325 °C/40 h)	0	2	no
Ni4N5_001b	Ni4N5	IFrun006	0	2	no
Ni4N5_010	Ni4N5	IFrun016 (1200 °C)	0	4	no
Ni4N5_012	Ni4N5	IFrun019 (1200 °C)	0	1	no
Ni4N5_034	Ni4N5	IFrun049 (1200 °C/12 h)	4	0	yes
Ni4N5_053	Ni4N5	IFrun059 (1200 °C/12 h)	3	3	no
Ni4N5_081	Ni4N5	IFrun082 (1300 °C/20 h)	10	10	yes

Table 2: Specimens with committed characterization data and a matched, parsed furnace run. “SEM”/“Opt.” are committed file counts; “EBSD” flags committed inverse-pole-figure/OIM data. Full inventory in paper/characterization-crossref.csv.

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Declaration of competing interests. **[TODO:]** — provisional: the authors declare no competing interests.

Data availability. design files, logs, and extracted context are in the development repository (vertical-cloud-lab/custom-induction-furnace, <https://github.com/vertical-cloud-lab/custom-induction-furnace>);

an archival snapshot is deposited at Zenodo, <https://doi.org/10.5281/zenodo.20878017>.

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Hazard	Failure mode	Consequence	Mitigation
High voltage / RF field	Contact with coil or induced currents in nearby conductors	Electric shock, burns	Keep conductive objects/electronics $\geq 6-12$ in from the coil while powered; isolate DAQ/control path
Lockout-tagout	Internal transformer-tap change (LEPEL prototype)	Shock during service	Follow LOTO before opening the generator
Vacuum / quartz	Venting against a spinning turbo pump; quartz fracture/implosion	Pump destruction; flying glass	Never vent while the turbo spins; handle quartz/gauge carefully; use eye protection
Water + electricity	Coil/chiller leak near energized hardware	Shock, equipment damage	Confirm flow and check for leaks before RF; route water away from electronics
Thermal	Sample melt-through; hot graphite oxidizing on vent	Pump-line contamination, burns	Cool under vacuum; let cool before venting; nitrile gloves when loading/unloading
Inert gas	Backfill/venting in an enclosed space	Asphyxiation	Adequate ventilation; do not backfill in confined areas
EMI	RF coupling into nearby instruments	Data corruption, instrument damage	Shield/separate sensitive electronics; keep them clear of the coil

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